

Practice Set: Constant Acceleration Kinematics

This practice set covers **one-dimensional motion with constant (uniform) acceleration**. For each problem, try to solve it yourself first, then expand the solution to check your work.

Formula Reference

Let x_0 and v_0 be the position and velocity at time $t = 0$, and x, v be the position and velocity at a later time t . For constant acceleration a :

$$v = v_0 + at$$

$$x = x_0 + v_0t + \frac{1}{2}at^2$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

$$x = x_0 + \frac{v_0 + v}{2}t$$

Sign convention: choose one direction as positive and stay consistent for the whole problem. For free fall near Earth's surface, use $a = -9.8 \text{ m/s}^2$ if "up" is positive (or $g = 9.8 \text{ m/s}^2$ downward if "down" is positive).

Problem 1: Reading a Velocity-Time Graph

A cart moves along a straight track. Its velocity-time graph is a straight line that starts at $v = 2 \text{ m/s}$ when $t = 0 \text{ s}$ and rises to $v = 10 \text{ m/s}$ at $t = 4 \text{ s}$.

What is the cart's acceleration, and what does the *area* under this graph represent?

► Solution

Problem 2: Basic Velocity Update

A car starts from rest and accelerates at 3 m/s^2 . What is its velocity after 5 seconds?

► Solution

Problem 3: Displacement from Rest

Using the same car from Problem 2 (starts from rest, $a = 3 \text{ m/s}^2$), how far does it travel in the first 5 seconds?

► Solution

Problem 4: Finding Acceleration from Velocity and Distance

A bicycle traveling at 4 m/s speeds up uniformly and reaches 12 m/s after covering 32 meters. What was its acceleration?

► Solution

Problem 5: Braking Distance

A car traveling at 20 m/s brakes with a constant deceleration of 5 m/s^2 . How far does it travel before coming to a stop?

► Solution

Problem 6: Time to Stop

Using the same car from Problem 5 (initial speed 20 m/s , deceleration 5 m/s^2), how long does it take to stop?

► Solution

Problem 7: Free Fall Drop

A ball is dropped (initial velocity 0) from a height of 45 meters. Using $g = 9.8 \text{ m/s}^2$, how long does it take to hit the ground, and how fast is it moving when it lands?

► Solution

Problem 8: Ball Thrown Upward

A ball is thrown straight up with an initial velocity of 19.6 m/s . Using $g = 9.8 \text{ m/s}^2$, find (a) the maximum height it reaches and (b) the total time until it returns to the launch height.

► Solution

Problem 9: Reaction Time + Braking (Two-Stage Motion)

A driver traveling at 24 m/s takes 0.75 s to react before braking (during reaction time, velocity stays constant). Once braking begins, the car decelerates at 6 m/s^2 until it stops. Find the **total** stopping distance, including the reaction-time distance.

► Solution

Problem 10: Conceptual — Same Acceleration, Different Interpretation

Two objects, A and B, both have a constant acceleration of -4 m/s^2 . Object A starts at $v_0 = 10 \text{ m/s}$ (moving forward) and object B starts at $v_0 = -10 \text{ m/s}$ (moving backward). Without calculating any numbers, describe what each object is doing (speeding up or slowing down) over the next few seconds, and explain your reasoning.

► Solution

Answer Key

#	Answer
1	$a = 2 \text{ m/s}^2$; area under $v-t$ graph = displacement
2	$v = 15 \text{ m/s}$

#	Answer
3	$x = 37.5 \text{ m}$
4	$a = 2 \text{ m/s}^2$
5	40 m
6	$t = 4 \text{ s}$
7	$t \approx 3.03 \text{ s}, v \approx 29.7 \text{ m/s}$
8	max height = 19.6 m, total time = 4 s
9	66 m
10	A slows down then reverses; B speeds up backward